

CS-540 Data Literacy & Visualizations

10-1 Journal: Visualizing Your Journey

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Throughout this course, I have developed a much deeper understanding of how data can be transformed into meaningful insights that support decision making. At the beginning of the course, I viewed data primarily as numbers and outputs, but over time, I learned how to analyze, interpret, and present data in a way that tells a clear and impactful story. This experience has strengthened both my technical skills and my ability to think critically about how data is used in real-world contexts.

One of the most challenging aspects of this coursework was learning how to properly clean, integrate, and prepare multiple datasets for analysis. At first, combining data from different sources felt overwhelming because of inconsistencies in formatting, missing values, and differences in structure. For example, when working on my final project, I had to merge transit ridership data with population data. Ensuring that the datasets aligned correctly requires careful preprocessing, including standardizing text fields, converting data types, and verifying that key columns matched. I overcame this challenge by breaking the process into smaller steps, using exploratory data analysis techniques to understand the structure of the data, and applying methods such as data cleaning and merging systematically. Over time, I became more confident in finding and resolving these issues, which significantly improved the quality of my analysis.

A key data literacy skill that I developed in this course is the ability to create meaningful data visualizations that communicate insights effectively. This is a skill I plan to apply to in both academic and professional settings. In my future career, particularly in roles related to data

analysis or software development, I will need to present complex information to different audiences, including both technical and non-technical stakeholders. The ability to design clear and purposeful visualizations will allow me to communicate findings more effectively and support data-driven decision making. For example, in my final project, I used both a high-level bar chart and a detailed scatter plot to present different perspectives of the same data. This approach proved how visualizations can be tailored to meet the needs of different audiences, which is a critical skill in professional environments.

My use of data visualizations played a significant role in supporting a data-driven narrative in my final project. The high-level bar chart provided a clear overview of transit ridership across different agencies, making it easy for executive stakeholders to quickly show trends and differences. In contrast, the detailed scatter plot incorporated multiple variables, including population, ridership, and running expenses, allowing for a more in-depth analysis. This combination of visualizations created a layered narrative that guided the audience from a general understanding to more specific insights. By structuring the visualizations in this way, I was able to tell a cohesive story that highlighted key patterns in the data and explained their significance. This experience showed me how powerful visualizations can be when they are intentionally designed to support a narrative rather than simply display information.

My perspective on working with data has changed significantly since the beginning of the course. Initially, I focused mainly on performing calculations and producing outputs, without fully considering how the data would be interpreted or used. Now, I understand that working with data is not just about analysis, but also about communication, context, and decision making. I have learned that the way data is presented can greatly influence how it is understood, and that effective data work requires both technical accuracy and thoughtful design. I also gained a

greater appreciation for the importance of data quality, as even small inconsistencies can affect the validity of results. Overall, I now approach data with a more analytical and strategic mindset, focusing not only on what the data shows, but also on what it means and how it can be used to inform decisions.

In conclusion, this course has provided me with valuable skills in data analysis, visualization, and storytelling that I will carry forward into future academic and professional work. I have grown more confident in handling complex datasets, creating meaningful visualizations, and communicating insights clearly. These experiences have prepared me to use data more effectively and to approach problems with a more structured and informed perspective.